

TARUNI NALLAMOTHU

taruninallamothu24@gmail.com | +44 7741031780 | London, UK | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

AI/ML Engineer and Data Engineer with over three years of industry experience building machine learning, Generative AI, NLP and data solutions using Python, LLMs, RAG, Azure, Databricks and cloud-native engineering. Hands-on experience with agentic and multi-agent AI workflows, semantic search, conversational AI, transformer models, REST APIs, microservices, model serving, MLOps and scalable ETL/ELT pipelines. Experienced with Azure OpenAI, Azure AI Foundry, Azure Machine Learning, MLflow, Docker, Kubernetes, CI/CD, Redis, monitoring and secure enterprise AI patterns, alongside a strong foundation in PySpark, SQL, statistics and predictive modelling.

TECHNICAL SKILLS

Programming & Backend: Python, FastAPI, Flask, REST APIs, Microservices Architecture, SQL/T-SQL, PySpark

AI/ML & Generative AI: Generative AI, Large Language Models (LLMs), RAG, Agentic AI, Multi-Agent AI Systems, Conversational AI, NLP, Prompt Engineering, Semantic Search, Embeddings, Transformer Fine-Tuning, BERT, DistilBERT, RoBERTa, Hugging Face Transformers, PyTorch, TensorFlow, Scikit-learn, XGBoost, SHAP

Cloud & AI Platforms: Azure OpenAI, Azure AI Foundry, Azure Machine Learning, Azure AI/Cognitive Search, Azure Databricks, MLflow, Azure Data Factory, Synapse Analytics, Azure SQL

MLOps & Infrastructure: Docker, Docker Compose, Kubernetes, Azure DevOps, GitHub Actions, CI/CD, Model Serving, Inference Pipelines, Model Versioning, Redis, Kafka

Monitoring & Security: Azure Monitor, Application Insights, Observability, Distributed Tracing, RBAC, Azure Active Directory

Data Engineering & Storage: Apache Spark, Delta Lake, ETL/ELT, Medallion Architecture, Airflow, Data Warehousing, Dimensional Modelling, SQL Server, MongoDB, MySQL, Delta Tables, ChromaDB, Vector Search

CERTIFICATIONS

- Google Certified: Associate Cloud Engineer
- Microsoft Certified: Azure Data Scientist Associate

PROFESSIONAL EXPERIENCE

Machine Learning Engineer (Intern, Remote) | Hansvonlehwalte

Jan 2026 – Present

- Designed and developed Generative AI and conversational AI solutions using Python, FastAPI, LLMs and RAG for enterprise knowledge retrieval, support-ticket intelligence and workflow automation, cutting average workflow processing time from 12 minutes to 7 minutes.
- Built RAG pipelines using embeddings, semantic search and vector stores to ground LLM responses in enterprise knowledge sources, improving retrieval quality from 70% to 86%.
- Developed agentic and multi-agent AI workflows for multi-step tasks, coordinating retrieval, reasoning and API/tool execution across specialized AI components.
- Built REST APIs and microservices with Python and FastAPI to expose ML and LLM capabilities to downstream applications and frontend systems, reducing average API response latency from 4 seconds to under 3 seconds.
- Implemented conversational memory and session-context management using Redis, maintaining 24-hour session context to support multi-turn AI interactions and context-aware responses.
- Applied Azure OpenAI and Azure AI Foundry capabilities for LLM applications, prompt engineering, embeddings and enterprise AI integration.
- Used MLflow for experiment tracking, model versioning and lifecycle management; containerized AI services with Docker and Kubernetes and supported CI/CD through Azure DevOps and GitHub Actions, cutting CI/CD pipeline time from 45 minutes to 25 minutes.
- Implemented monitoring and observability with Azure Monitor, Application Insights and structured logging, and applied Azure AD/RBAC controls to secure APIs and AI services, reducing average troubleshooting time from 60 minutes to 40 minutes.

Systems Engineer | Tata Consultancy Services, India

Dec 2021 – Aug 2024

- Engineered scalable data and ML pipelines using Azure Databricks, PySpark, Python, SQL and Azure Data Factory, processing 500GB+ of enterprise data daily from multiple source systems into curated analytics layers, improving pipeline execution efficiency by 35% and reducing execution time by 40%.
- Built reusable cleansing and validation frameworks across 1M+ records, enforcing schema, completeness, duplicate and business-rule checks before downstream warehouse and analytics consumption, achieving 99.9% data accuracy.
- Created PySpark/Python feature-engineering pipelines generating 100+ model-ready variables for classification, forecasting, anomaly-detection and predictive analytics workloads, raising anomaly-detection accuracy to 95% and improving a classification model's AUC from 0.71 to 0.86.
- Built and evaluated machine learning and NLP solutions using Python, Scikit-learn and transformer-based approaches for classification, predictive analytics and enterprise text-processing use cases, improving forecasting accuracy from 78% to 92% through refined feature engineering and model tuning.
- Developed semantic-search and knowledge-retrieval workflows using embeddings, transformer-based NLP and vector-search techniques for large enterprise document collections.

- Developed Python REST APIs and microservices to expose data and ML functionality to downstream enterprise applications and supported model-serving and inference workflows.
- Used Azure Machine Learning and MLflow for experiment tracking, model versioning, evaluation and management of ML workloads across development environments.
- Containerized application and ML components using Docker and Kubernetes and automated testing, deployment and version management using Azure DevOps and GitHub-based CI/CD workflows.
- Implemented monitoring and troubleshooting using Azure Monitor, Application Insights and centralized logging, identifying data-quality, API, infrastructure and application failures and improving system uptime by 20% through proactive root-cause analysis.
- Developed dimensional datasets and reporting-ready data marts supporting 20+ operational and financial KPIs; optimized T-SQL stored procedures, views and reusable transformations for reliable downstream reporting, cutting manual reporting effort by 50%.

IT Consultant | Andhra Pradesh Capital Region Development Authority (APCRDA)

Oct 2019 – Nov 2021

- Supported civil engineering and public infrastructure projects by collecting, organizing and validating field survey, geospatial and project data used for planning, progress tracking and technical reporting.
- Managed engineering datasets covering survey measurements, project locations, quantities, infrastructure assets and construction progress, ensuring accuracy and consistency across project records.
- Used Excel, basic SQL and data-management techniques to structure, validate and analyze engineering data, identifying missing, duplicate and inconsistent records.
- Prepared project progress reports, engineering summaries and data-driven documentation by consolidating information from field engineers, survey teams and project stakeholders.
- Developed foundational experience in databases, structured data, data validation and digital information management, which supported the subsequent transition into IT and data engineering.

PROJECT

NovaBank – AI Customer Support Intelligence

Python, PyTorch, DistilBERT, XGBoost, SHAP, RAG, Sentence Transformers, ChromaDB, Gemini, FastAPI, Next.js, Docker

- Built an end-to-end AI-powered customer support platform combining NLP, predictive machine learning and Generative AI through a RAG pipeline to analyze support tickets, retrieve grounded knowledge and assist operational decision-making.
- Fine-tuned a DistilBERT intent classifier to 90.59% F1 and developed an XGBoost SLA-breach predictor achieving 77.98% recall, with SHAP-based explainability for model interpretation.
- Integrated Sentence Transformer embeddings and ChromaDB vector search with Gemini for retrieval-grounded response generation, reducing dependence on ungrounded LLM responses.
- Exposed AI capabilities through a FastAPI backend, integrated a Next.js frontend and containerized the application with Docker for reproducible deployment.

VolatilityIQ – Cryptocurrency Volatility Forecasting

Python, GARCH, Random Forest, XGBoost, LSTM (TensorFlow/Keras), Sentiment Analysis

- Benchmarked GARCH, a time-series volatility model, against Random Forest, XGBoost and LSTM to forecast volatility across ten cryptocurrencies, combining price-based and sentiment-derived features.
- Applied a time-aware train-test split to avoid look-ahead bias, evaluating predictive accuracy, robustness and interpretability across models rather than reporting a single result.
- Completed as the final project for the MSc in Data Science, applying quantitative finance and machine learning techniques to real cryptocurrency market data.

Medical Text Analyzer – Healthcare NLP & Predictive Analytics

Python, NLTK, TF-IDF, SVM, Random Forest, TextRank, Scikit-learn, Streamlit

- Developed an end-to-end medical NLP application to process unstructured clinical text, generate concise document summaries and predict patient length-of-stay categories through an interactive Streamlit interface.
- Built an NLP preprocessing pipeline using NLTK, tokenization, stop-word handling and lemmatization, preserving medically meaningful terms while transforming clinical text into model-ready features.
- Engineered TF-IDF text features and trained/evaluated SVM and Random Forest classifiers for medical-text classification, including stratified train/test splitting and classification-performance evaluation.
- Implemented TextRank-based extractive summarization alongside configurable compression and model-confidence visualizations, combining NLP summarization and predictive analytics in a single healthcare application.

EDUCATION

Master of Science in Data Science | University of Hertfordshire | GPA: 3.91/4.0 | Sep 2024 – Oct 2026

Student Representative | University of Hertfordshire | Nov 2024 – Oct 2026

Bachelor of Technology in Civil Engineering | V.R Siddhartha Engineering College, JNTUK | CGPA: 8.17/10.0